

ANDROID APP DEVELOPMENT USING GENAI – JAL SANCHAY TRACKER

Introduction

Jal-Sanchay Tracker is an Android application developed to promote water conservation by helping users calculate and monitor rainwater harvesting. The application provides an easy way to estimate the amount of rainwater collected based on roof area and rainfall data. It also tracks total water savings and displays historical records for better awareness and analysis.

Problem Statement

Many households and small communities are unable to estimate or track the amount of rainwater that can be harvested during rainfall. Due to the absence of simple monitoring systems, a large amount of water is wasted. This project aims to provide a digital solution that encourages water conservation through accurate rainwater harvesting calculations and tracking.

OBJECTIVES

- To calculate the amount of rainwater collected.
- To track total water savings over time.
- To maintain a history of calculations.
- To provide a simple and user-friendly Android application.
- To spread awareness about water conservation.

Features

- User enters roof area and rainfall values.
- Application calculates collected rainwater.
- Displays total water savings.
- Stores and shows calculation history.
- Monthly water-saving report display.
- Water efficiency progress indicator.
- Reset option to clear data.
- Attractive and responsive user interface.

Technologies Used

- Android Studio
- Kotlin
- XML
- Gradle Build System

Frontend

- XML layouts for user interface
- Buttons, TextViews, EditTexts, and ProgressBars
- Responsive UI design with colors and styling

Backend

- Kotlin logic for water calculation
- Event handling using button click listeners
- Data processing and result generation

Working Process

- User enters roof area.
- User enters rainfall in millimeters.
- App calculates water collected using formula:
- $\text{Water Collected} = \text{Area} \times \text{Rainfall} \times 0.0929 \times 0.8$
- Result is displayed on screen.
- Total savings and history are updated.
- Progress bar shows efficiency level.

Expected Outcome

The application helps users understand the benefits of rainwater harvesting and encourages efficient water management by providing instant calculations and tracking features.

FUTURE ENHANCEMENT

- Integration with cloud database for permanent data storage.
- Graphs and charts for better water usage analysis.
- User login and profile management system.
- Weather API integration for automatic rainfall updates.
- Exporting reports in PDF format.
- Notifications and alerts for water-saving goals.
- Multi-language support for regional users.
- AI-based prediction for future water collection.

Conclusion

Jal-Sanchay Tracker is a simple yet effective Android application designed to support water conservation efforts. The project demonstrates Android development concepts including frontend UI design, backend logic implementation, and user

interaction handling while addressing an important environmental issue.